GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering

Level: Diploma

Branch: Chemical Engineering

Subject Code: DI05005021

Subject Name: Chemical Engineering Plant Economics

w. e. f. Academic Year:	2026-2027
Semester:	V
Category of the Course:	PCC

Prerequisite:	Students should have basic knowledge of mathematics, material and energy balance, and fundamental chemical engineering concepts like unit operations and process flow diagrams. Understanding of equipment such as reactors, pumps, boilers, distillation column, heat exchangers etc. is helpful for estimating utility and maintenance costs, basic financial concepts for economic analysis. These prerequisites help students understand cost estimation, profitability analysis, and investment decisions in chemical plants.
Rationale:	A plant-design project moves to completion through a series of stages starting from preliminary evaluation of economics and market to commercial production. Chemical engineering design of a new chemical plant and the expansion or revision of existing one require the use of engineering principles and theories combined with consideration of practical limits imposed by industrial conditions. In this course special emphasis is given on the applied economics and engineering principles involved in the design of chemical plants. Use of these principles is highly required for any successful chemical engineer to work in the area of production, administration, sales, marketing, research, and development of a new chemical project..

Course Outcome:

After Completion of the Course, Student will able to:

No	Course Outcomes	RBT Level
01	Gain basic knowledge of chemical engineering plant and process design in industries.	R, U
02	Understand a process to start new project by considering the concept of plant and process design.	R, U
03	Select process equipment or instruments of the same function based on both technical and commercial point of view.	U, A
04	Choose appropriate plant location and plant layout for the chemical project.	U, A
05	Apply knowledge of economic for project to run an industry with profitability.	R, U, A

**Revised Bloom's Taxonomy (RBT)*

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Teaching and Examination Scheme:

Teaching Scheme (in Hours)			Total Credits L+T+ (PR/2)	Assessment Pattern and Marks				Total Marks
				Theory		Tutorial / Practical		
L	T	PR	C	ESE(E)	PA(M)	PA(I)	ESE (V)	
3	1	0	4	70	30	0	0	100

Course Content:

Unit No.	Content	No. of Hours	% of Weightage
Unit – I **Introduction**	1.1 Role of Chemical Engineer. 1.2 Chemical Engineering Design. 1.3 Need of plant design. 1.4 Process Design 1.5 Design and selection of chemical engineering equipment 1.6 Criteria for good design	06	14%
Unit– II Development of project for plant and process design	2.1 Chemical engineering plant project objective. 2.2 Process evolution stages and their Importance. 2.3 Plant design factors 2.3.1 Technical factors 2.3.2 Economic factors 2.3.3 Legal factors 2.3.4 Safety and sanitation. 2.4 Source of Information. 2.5 Continuous v/s Batch processing, 2.6 Shift and Operating schedules 2.7 Types of flow diagrams.	09	20%

Unit– III **Selection of chemical engineering equipments**	3.1 selection of material 3.2 Plan for selection of equipment 3.3 Selection of process equipment and specification sheet for equipment 3.4 Standard v/s Special equipment. 3.5 Selection of equipment (a) Size reduction equipment, (b) Heat transfer equipment, (c) Material handling equipment (d) Mass transfer equipment (e) Pumps 3.6 Piping, Pipe strength and wall thickness 3.7 Piping design problems, 3.8 Types of insulation, Factors governing selection of insulation.	10	22%
Unit– IV **Plant Layout and Location**	4.1 Factors of plant layout 4.2 Principles of plant layout 4.3 Factors for selection of plant location: Primary factors and specific factors	05	11%
Unit– V **Economic evaluation of the project**	5.1 Capital investment. 5.2 Fixed capital investment, 5.3 Working capital investment. 5.4 Factors affecting investment and production costs. 5.5 Cost Indices, Cost-Size relation, and cost-Time relation. 5.6 Numerical based on Cost Indices 5.7 Depreciation and it's types 5.8 Methods for determining depreciations Arbitrary methods, Methods with interest on investment 5.8.1 Numeric based on depreciation. 5.9 Total product cost (TPC) 5.10 Profitability analysis: Net and gross earnings 5.11 Percent return on investment, Pay-out time period, Present worth, Turn-over ratio 5.12 Break-even analysis (Analytical method), 5.13 Break-even chart (Graphical method), 5.14 Numerical of Break-even analysis	15	33%
	Total	**45**	**100%**

Suggested Specification Table with Marks (Theory):

Distribution of Theory Marks (in %)					
R Level	**U Level**	**A Level**	**N Level**	**E Level**	**C Level**
20%	**45%**	**35%**	---------	————————	————————

Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)

References/Suggested Learning Resources:

(a) Books:

S. No.	Title of Book	Author	Publication with place, year and ISBN
1	Chemical Engineering Plant Design.	Vilbrandt, Frank Carl and Dryden,Charles E.	McGraw Hill, New Delhi, 4th edition
2	Plant Design and Economics for Chemical Engineers	Peters, Max and Klaus Timmerhaus	McGraw Hill, New Delhi, 4th edition
3	Chemical engineering Design Principles, Practice and Economics of Plant and Process Design	Gavin towler Ray sinnott	Butterworth-Heinemann Elsevier (2008)
4	Process Engineering Economics	Couper, James R.	Marcel and Dekker

(b) Open-source software and website:

- https://nptel.ac.in
- www.cheresources.com
- http://people.clarkson.edu/~wwilcox/Design/refcosts.html
- http://app.knovel.com/web/toc.v/cid:kpCEDPPEP4
- https://www.lib.utexas.edu/chem/info/chemengecon.html
- http://www.mhhe.com/engcs/chemical/peters/data/ce.html

Suggested Tutorial List:

Sr. No.	Tutorial (Course Outcomes in Psychomotor Domain according to NBA Terminology)	Approx. Hrs Required
1	Prepare block type and equipment flow diagram for production of desired (quantity and quality) chemical	1
2	Prepare block type material balance flow diagram for production of desired (quantity and quality) chemical	2

3	Prepare block type energy balance flow diagram for production of desired (quantity and quality) chemical	2
4	Prepare detailed process and instrumentation flow diagram for production of desired (quantity and quality) chemical	2
5	Prepare specification sheet for 1-2 shell and tube heat exchanger	1
6	Prepare specification sheet for packed type distillation column	1
7	Solve given simple problems using cost–size relationship and cost- time relationship (Cost indices)	1
8	Calculate depreciation using Straight line method for given plant	1
9	Calculate depreciation using Declining balance method for given plant	1
10	Calculate depreciation using Sum of the years digits method for given plant	1
11	Calculate depreciation using Sinking fund method for given plant	1
12	Find-out break-even point Using Analytical and Graphical methods for given plant.	2

Suggested Activities for Students:

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should perform following activities in group and prepare reports of about 5 pages for each activity. They should also collect/record physical evidences for their (student's) portfolio which may be useful for their placement interviews:

Following is the list of proposed student activities like:

1. Assignments
2. Technical Quiz/MCQ Test
3. Presentation on some course topic
4. I-net based assignments
5. Undertake micro-Project in team/individually

- Prepare chart/model of economic evolution or cost estimation of given project.
- Prepare chart of process design.
- Prepare chart/model types flow diagram.
- Prepare specification sheet of equipment of given project.
- Draw suitable chart for plant design.
- Prepare 15-20 slides power point presentation on any topic of project engineering.

- Prepare material balance diagram of given project.
- Prepare detailed IPD (instrumentation and process diagram) of given project.
- Prepare plant layout of given project.
- Prepare block flow diagram of given project.

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